**Table: 10.1** Ionic Channels in the Axolemma

| | |
|---|---|
| **Leakage Channels** are ionic channels that remain open all the time | $K^+$ leakage channels are more in number than the $Na^+$ leakage channels. Sarcolemma has greater permeability to $K^+$ ions than $Na^+$ ions. These ions keep moving continuosly maintain the potential difference across the axolemma. |
| **Ligand-gated Channels** are chemically gated channels which open or close in response to a chemical stimuli. | They are located between the presynaptic membrane of the first axon and post synaptic membrane of the cell body of second neuron [i.e. dendrites and cell bodies]. The neurotransmitter acetylcholine opens ligand channels that allow $Na^+$ and $Ca^{++}$ ions diffuse inward and $K^+$ ions diffuse outward. |
| **Voltage-gated Channels** are mechanically gated channels which open in response to a physical stimulus in the form of vibration such as touch and pressure. | These channels open in response to a change in membrane potential.<br>There are two types of voltage-gated channels.<br>**i. Sodium voltage-gated channel**<br>**ii. Potassium voltage-gated channel** |

$Na^+$ influx into the cell. When the axon is not conducting any impulses i.e. in resting condition, the axon membrane is more permeable to $K^+$ and less permeable to $Na^+$ ions, whereas it remains impermeable to negatively charged protein ions.

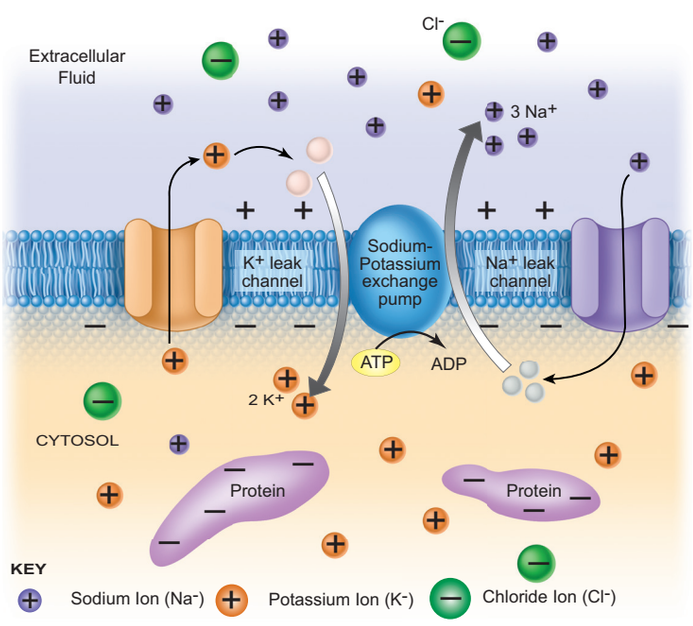



**Figure 10.3** Ionic channels

The axoplasm contains high concentration of $K^+$ and negatively charged proteins and low concentration of $Na^+$ ions. In contrast, fluid outside the axon (ECF) contains low concentration of $K^+$ and high concentration of $Na^+$, and this forms a concentration gradient. This ionic gradient across the resting membrane is maintained by ATP driven **Sodium-Potassium** pump, which exchanges $3Na^+$outwards for $2K^+$ into the cells. In this state, the cell membrane is said to be **polarized**. In neuron, the **resting membrane potential** ranges from **-40mV** to **-90mV**, and its normal value is **-70mV**. The minus sign indicates that the inside of the cell is negative with respect to the outside (Figure 10.4).



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